M1 Residue-Line Tools and Conjecture F Reductions

RS-MCA experimental note
integrated from AllenGrahamHart proof-program PRs, with Codex editing

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Abstract

This note consolidates the M1 and Conjecture F material that may be useful for a future CAP25 revision. The results here do not close the M1 safe-side residue-line local limit. They do, however, isolate several reusable theorem blocks: exact first and second moments for split-locator alignment in the random-word model; reductions removing common divisors and quotient-pullbacks from Conjecture F; fixed-dimensional and dimension-two incidence bounds; quotient seam arithmetic; subgroup Hankel displacement identities; Johnson exchange-mixing bounds; and a deficiency-one SPI eliminant-or-residual chart.

The intended use in Paper D/CAP25 is as a residual-branch toolkit. Paid tangent and quotient cells are separated first. What remains is the aperiodic residue-line branch. This note says which parts of that branch are already formal algebra/combinatorics and which parts remain genuine inverse or incidence theorems.

1 Status and Interface with CAP25

The CAP25 safe-side program asks for upper bounds on bad line parameters after known tangent, quotient, and extension cells have been charged. In the notation of exact agreement A,

$$j = n - A, \qquad t = A - k, \qquad j + t = n - k.$$

The M1 problem is to control aperiodic residue-line packings in this (j, t)-window.

This note has three status levels:

(i) proved local algebra/combinatorics, such as the first-moment, Conjecture F reductions, quotient seam, displacement identities, Johnson mixing, and deficiency-one chart;

(ii) experimental toy evidence, such as small-field Conjecture F censuses;

(iii) open inverse/classification steps, such as the full M1 local limit, XR inverse theorem, high-dimensional Conjecture F, and identically valid SPI residuals.

2 Split-Locator Moment Calculus

Let $F = \mathbb{F}_q$, let $D \subset F^*$ have size n, and let $k < A \le n$. Put

$$t = A - k, \qquad j = n - A.$$

For $R \subset D$ with $|R| = j$, define

$$\ell_R(X) = \prod_{r \in R} (X - r).$$

For a word $w : D \to F$, define the locator-syndrome vector

$$S_R(w) = \left(\sum_{x \in D} w(x) \ell_R(x) x^m \right)_{m=1}^{t} \in F^t.$$

For independent uniform words $u, v : D \to F$, call R *aligned* if $S_R(v) \neq 0$ and $S_R(u) \in F S_R(v)$.

Theorem 2.1 (exact first moment). *The expected number $N_A(u, v)$ of aligned split locators is*

$$\mathbb{E} N_A = \binom{n}{j} (1 - q^{-t}) q^{1-t}.$$

Proof. For fixed R, the map $w \mapsto S_R(w)$ is surjective. Indeed, on $E = D \setminus R$ its matrix has entries $\ell_R(x) x^m$; after nonzero column scaling it is a $t \times A$ Vandermonde block of rank t. Thus $(S_R(u), S_R(v))$ is uniform on $F^t \times F^t$. The number of pairs (a, b) with $b \neq 0$ and $a \in Fb$ is $(q^t - 1)q$, giving probability $(1 - q^{-t}) q^{1-t}$. Sum over $\binom{n}{j}$ locators. □

Theorem 2.2 (two-locator joint rank). *For $R, T \subset D$ of size j, put $c = |R \cap T|$. The map*

$$w \mapsto (S_R(w), S_T(w))$$

has rank

$$2t - \max(0, t - j + c).$$

Proof. A linear relation between the two syndrome blocks has the form

$$\ell_R A + \ell_T B = 0, \qquad \deg A, \deg B \leq t - 1.$$

Writing $G = \gcd(\ell_R, \ell_T)$, $\ell_R = Gr$, $\ell_T = Gs$, with $\deg G = c$ and $(r, s) = 1$, gives $rA + sB = 0$. Hence $A = sH$ and $B = -rH$ with

$$\deg H \leq t - 1 - (j - c).$$

The relation space therefore has dimension $\max(0, t - j + c)$. □

Corollary 2.3 (covariance support). *The alignment indicators for R and T are pairwise independent whenever*

$$d(R, T) := j - |R \cap T| \geq t.$$

All covariance is supported in the radius-$(t-1)$ Johnson neighborhood.

Theorem 2.4 (exact second moment). *Put*

$$h(c) = \max(0, t - j + c).$$

For $0 \leq h \leq t$, define

$$P_h = \frac{q(q^h - 1) q^{2(t-h)} + q^2 (q^{t-h} - 1)^2}{q^{4t-2h}}.$$

Then

$$\mathbb{E} N_A^2 = \sum_c \binom{n}{j} \binom{j}{c} \binom{n-j}{j-c} P_{h(c)},$$

where the sum ranges over valid overlaps c.

Proof sketch. Theorem 2.2 puts the joint image into a standard fiber product $U_h = \{(z, p; z, q)\}$. If the common part of $S_R(v), S_T(v)$ is nonzero, the two alignment scalars coincide; if it is zero, the two tail scalars are independent. Counting these two cases gives P_h, and summing over ordered pairs of locators gives the formula. □

Corollary 2.5 (dependency-degree concentration)**.** *Let*

$$D_t(n, j) = \sum_{d=0}^{\min(t-1,j,n-j)} \binom{j}{d}\binom{n-j}{d}.$$

Then

$$\mathrm{Var}(N_A) \le \mathbb{E}N_A \cdot D_t(n, j).$$

Consequently, if $\mathbb{E}N_A \ge n^{2(t-1)+s}$*, then*

$$\frac{\mathrm{Var}(N_A)}{(\mathbb{E}N_A)^2} \le n^{-s}.$$

Remark 2.6 (CAP25 meaning). FM1 is an average-over-random-word-pairs calculation. It shows that in the regular M3 window for the $F_{17^{32}}$, $n = 512$, $k = 256$ row, random aligned split locators are astronomically unlikely. It does not bound worst-case pairs; those require Conjecture F or exchange-rigidity input.

3 Conjecture F Reductions

Let K be a field, $H \subset K$ finite, and

$$P_H(X) = \prod_{h \in H}(X - h).$$

Let $\mathcal{D}_j(H)$ be the set of monic squarefree degree-j divisors of P_H. Equivalently, $\mathcal{D}_j(H)$ is the set of locator polynomials $L_S(X) = \prod_{h\in S}(X - h)$ for $|S| = j$.

Lemma 3.1 (common-GCD reduction)**.** *Let P be a projective linear space of polynomials of degree at most j, and let $E = P \cap \mathcal{D}_j(H)$. If every locator in E is divisible by a fixed monic divisor G of P_H, with* $\deg G = w$*, then division by G injects E into*

$$\mathcal{D}_{j-w}(H \setminus Z(G))$$

inside a projective linear space of dimension at most $\dim P$*.*

Lemma 3.2 (quotient-pullback recursion)**.** *Assume $H = \mu_n$ and* $\mathrm{char}\, K \nmid n$*. If* $M \mid \gcd(n, j)$*, then*

$$g(Y) \longmapsto g(X^M)$$

maps $\mathcal{D}_{j/M}(\mu_{n/M})$ bijectively onto the M-periodic stratum of $\mathcal{D}_j(\mu_n)$. Intersecting with a projective linear space descends to an intersection with a projective linear space of no larger dimension in the quotient locator space.

Lemma 3.3 (dimension-one voting)**.** *Let $P = \mathbb{P}(\mathrm{span}(L_0, L_1))$ be a projective line in $K[X]_{\le j}$, and assume no $h \in H$ is a common zero of L_0 and L_1. Then*

$$|P \cap \mathcal{D}_j(H)| \le \left\lfloor \frac{|H|}{j} \right\rfloor.$$

For an affine parameter $L_0 + zL_1$, one may add the point at infinity separately.

Proof. Each $h \in H$ votes for the unique projective parameter for which $aL_0(h) + bL_1(h) = 0$. Every degree-j locator receives exactly j votes. ☐

Lemma 3.4 (hyperplane concurrency)**.** *Let $W \le K[X]_{\le j}$ be nonzero and assume no $h \in H$ is a common root of all elements of W. For each h, let*

$$E_h = \{[L] \in \mathbb{P}(W) : L(h) = 0\}.$$

Then $P(W) \cap \mathcal{D}_j(H)$ is exactly the set of projective points lying on at least j of the hyperplanes E_h.

Theorem 3.5 (projective-plane pair bound)**.** *Under the hypotheses of Lemma 3.4, if* $\dim \mathbb{P}(W) = 2$*, then*

$$|\mathbb{P}(W) \cap \mathcal{D}_j(H)| \le \frac{\binom{|H|}{2}}{j-1}.$$

If the evaluation lines E_h are pairwise distinct, the sharper bound

$$|\mathbb{P}(W) \cap \mathcal{D}_j(H)| \le \frac{\binom{|H|}{2}}{\binom{j}{2}}$$

holds.

Proof sketch. Group coincident evaluation lines by multiplicity. A high-incidence point with total multiplicity at least j receives at least $j-1$ cross-pairs of roots unless all lines are simple, in which case it receives at least $\binom{j}{2}$ pairs. Distinct line pairs meet in at most one projective point, so the total number of charges is bounded by $\binom{|H|}{2}$. ☐

Theorem 3.6 (fixed-dimensional Conjecture F bound)**.** *Let $W \le K[X]_{\le j}$ have vector dimension $d+1$, and assume W is gcd-trivial on H. Then*

$$|\mathbb{P}(W) \cap \mathcal{D}_j(H)| \le \binom{|H|}{d}.$$

If C is the common root set of W on H, the general bound after removing common roots is

$$|\mathbb{P}(W) \cap \mathcal{D}_j(H)| \le \binom{|H| - |C|}{d}.$$

Proof. For a locator point $[L]$ with root set R, the space $W(-R)$ has dimension one. Hence some d roots of R already cut the projective space down to the point $[L]$. Choosing the first such d-subset injects the locator points into the d-subsets of H. The common-root version follows by division by the common divisor. ☐

Remark 3.7 (what remains open)*.* Theorem 3.6 shows that the primitive Conjecture F core is a dimension-growing problem. The small-field evidence packets find expected common-root, twin-line, sparse-dependence, and quotient-compatible structures, but they do not prove the growing-dimensional incidence theorem.

4 Quotient Seam Arithmetic

Lemma 4.1 (GAP-2 quotient seam)**.** *Let n, k, A be fixed and put $j = n - A$, $t = A - k$, $r = n - k$, so $j + t = r$. If $M \mid \gcd(n, k)$, then $M \mid r$, and hence*

$$M \mid j \iff M \mid t.$$

On such buckets all quotient parameters $n/M, k/M, A/M, j/M, t/M$ are integral. If $M \mid n$ and $M \mid j$ but $M \nmid k$, then the support descends but the dimension and window do not; this is a non-rate-preserving seam.

Proof. The implication follows from $r = j + t$ modulo M. The last statement follows from $A = n - j$ and $t = A - k$. $\square$

5 Subgroup Hankel and Exchange Tools

Let $H = \langle \alpha \rangle$ be a finite multiplicative subgroup of a field F, and let $V_m[x, r] = x^r$ for $0 \le r < m$.

Proposition 5.1 (Hankel factorization)**.** *For any word $w : H \to F$, the rectangular syndrome Hankel block satisfies*

$$H_{t,s}(w) = V_t(H)^T \operatorname{diag}(w(x)) V_s(H).$$

If $X = \{x_0, \ldots, x_{m-1}\}$ is a set of distinct nonzero points and

$$A_h(X) = V_X^T \operatorname{diag}(x^h : x \in X) V_X,$$

then

$$\det A_h(X) = \det(V_X)^2 \prod_{x \in X} x^h.$$

Proposition 5.2 (Johnson exchange mixing)**.** *Let $J(n, j)$ be the Johnson graph on j-subsets, with degree $d = j(n - j)$. If $\mathcal{A}$ has density δ and T_1 is obtained from uniform T_0 by one exchange, then*

$$\Pr[T_0, T_1 \in \mathcal{A}] \le \delta^2 + \left(1 - \frac{n}{d}\right) \delta(1 - \delta).$$

Point dictators attain equality.

Proof. This is the standard spectral bound for the normalized adjacency operator of $J(n, j)$. The first nontrivial eigenvalue is $1 - n/(j(n - j))$. $\square$

Proposition 5.3 (anticode packing)**.** *If a family $\mathcal{F}$ of j-subsets satisfies $|A \setminus B| > s$ for all distinct $A, B \in \mathcal{F}$, then*

$$|\mathcal{F}| \binom{j}{s} \le \binom{n}{j - s}.$$

Remark 5.4. These Johnson-scheme tools are support-combinatorial. They identify paid dictator/common-root shapes and packing losses, but cannot by themselves produce the finite-field q-scale value-set input needed for the prize corridor.

6 SPI Deficiency-One Chart

The SPI route studies one-parameter Hankel pencils in a deficiency-one chart. Let

$$M(Z) = M_0 + ZM_1$$

be a $t \times (j+1)$ matrix pencil with affine-linear entries, and assume $t = j$. For $0 \le i \le j$, let $M_i(Z)$ be the maximal minor after deleting column i and put

$$c_i(Z) = (-1)^i M_i(Z), \qquad L_Z(X) = \sum_{i=0}^{j} c_i(Z) X^i.$$

Theorem 6.1 (deficiency-one eliminant or residual)**.** *Assume H is the full root set of $X^n - 1$ and* $\operatorname{char} F \nmid n$*. Every finite slope of the deficiency-one chart lies in one of the following classes:*

(i) rank drop: all maximal minors vanish, bounded by the roots of a nonzero maximal minor of degree at most t;

(ii) low-degree full-rank chart: $c_j(z) = 0$, hence the Cramer locator has degree $< j$ and is charged to a contained higher-agreement branch;

(iii) top chart: $c_j(z) \neq 0$, and validity is equivalent to $L_z(X) \mid X^n - 1$.

On the top chart, pseudo-division gives

$$c_j(Z)^{\Delta}(X^n - 1) = Q(Z, X)L_Z(X) + R(Z, X), \qquad \Delta = n - j + 1,$$

with $\deg_Z R_m \le (n-j+1)t$ *for every remainder coefficient. Therefore either a nonzero remainder coefficient gives a finite eliminant of degree at most $t + (n-j+1)t$, or all remainder coefficients vanish identically and the family is a named valid residual branch.*

Remark 6.2. For $n = 512$, $k = 256$, $A = 384$, one has $t = j = 128$, hence the global one-parameter cap is

$$128 + 385 \cdot 128 = 49408.$$

This is a bounded eliminant problem unless the identically valid residual branch occurs. The theorem does not classify higher-dimensional SPI charts.

7 CAP25 Integration Checklist

A CAP25 v13-style M1 section can cite this note as follows:

(1) Use Lemmas 3.1, 3.2 and 4.1 to remove paid common-divisor and quotient-periodic strata before calling a branch aperiodic.

(2) Use Theorems 3.5 and 3.6 to dispose of fixed-dimensional and projective-plane Conjecture F subcases.

(3) Use Propositions 5.1 to 5.3 as the algebra and support-combinatorial input for XR or spectral routes, without claiming a full inverse theorem.

(4) Use Theorem 6.1 to state the first SPI chart as eliminant-or-residual, not as an unpriced black box.

(5) Treat FM1, Theorems 2.1 and 2.4, as random-pair evidence and a moment ledger, not a worst-case safe-side theorem.

8 Open Problems

Problem 8.1 (M1 local limit)**.** After tangent, quotient, extension, and fixed-dimensional cells are paid, prove a polynomial or target-level upper bound for all remaining aperiodic support-wise MCA residue-line parameters.

Problem 8.2 (Conjecture F in growing dimension)**.** Prove the primitive incidence bound for $\mathbb{P}(W) \cap \mathcal{D}_j(H)$ when $\dim \mathbb{P}(W)$ grows with n, after common divisors, quotient-pullbacks, twins, and sparse-dependence descents are removed.

Problem 8.3 (XR inverse theorem)**.** Classify near-extremizers of the Johnson exchange energy for aligned-locator families after paid common-root and quotient structures are stripped.

References

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